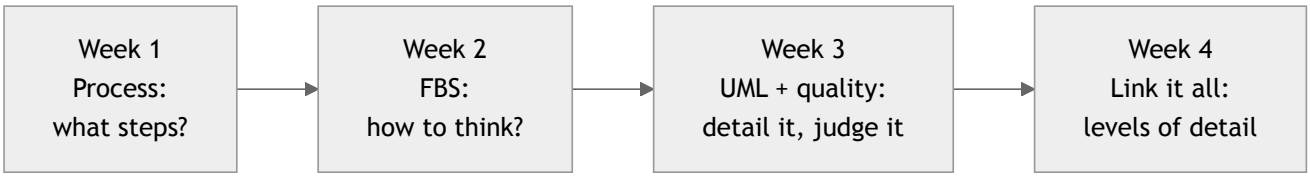
Software Conceptual Design: the story in 4 weeks

One story: **you must build a feature (Amazon Pay, then a mood-based music player). How do you go from idea to working software?**



Week 1: The process (what steps do we follow?)

1.1 Software = modules

Amazon is built from **modules** (separately addressable components, integrated to satisfy requirements).

Component	What it does
Inventory management	measures **amount, location, pricing, mix** of products; home page updates with **purchasing, seasonal trends, demand** (logistics + analytics); page is **personalised** by items viewed + shopping history
Payment gateway	authorises electronic payments (net banking, debit cards...); **middleman between bank and merchant site**. Flow: site sends **encrypted card info** → gateway **confirms with the bank** → money moves from user to merchant
Amazon Pay	a **mobile wallet** (digital cash): link card/bank, categories (recharge, bills, travel, rewards, gift vouchers), **add money**, **auto reload**

1.2 Five phases



Requirements. Goals the system must meet; they must cater to the clients. First step = understand the problem + study existing/similar systems (Paytm, PhonePe), **not** "learn a programming language". Think: **who** uses it, for **what**, in **what way**.

Client type	Example
External user	mobile banking app serves bank customers
Internal to the company	employee resource portal built by an internal team with HR
Another software	Razorpay gateway between shop site and bank

Design. Big-picture view; gives structure. Without it: developers have **different ideas** of how functionality works, change a function and **break a colleague's component**, delays, and **no overview to add new features**.

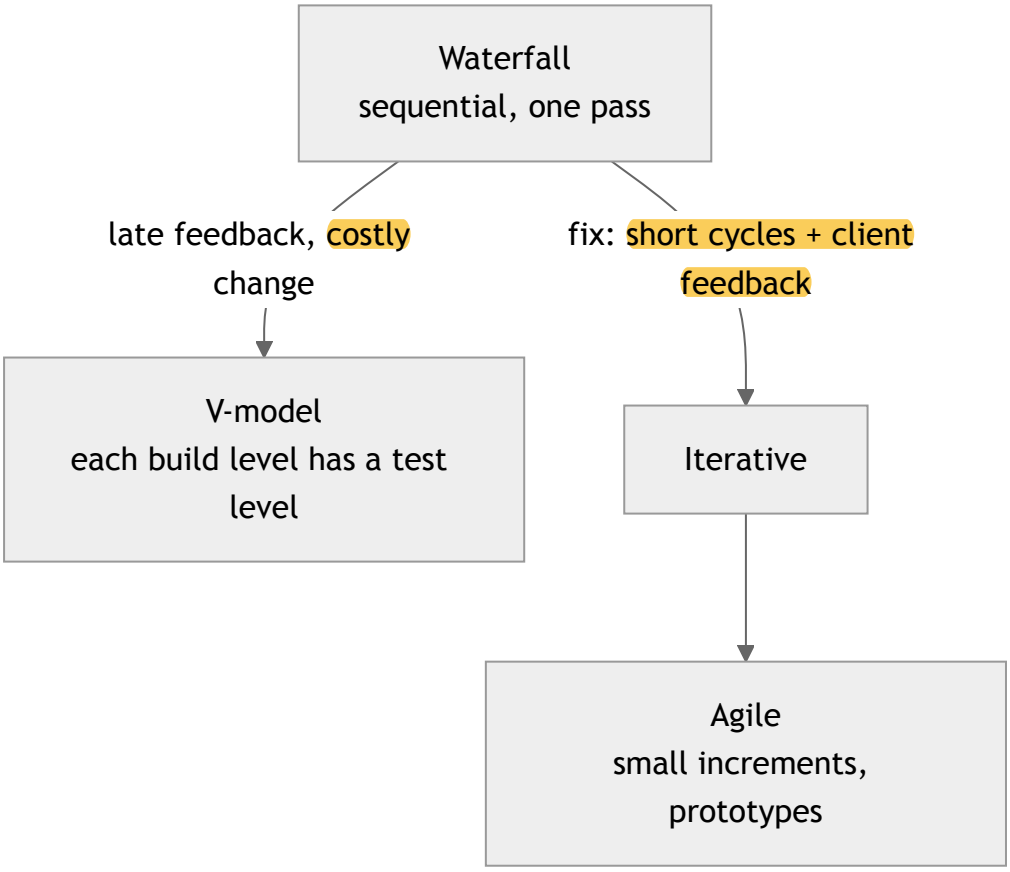
Development. Write code from requirements + design. **Distributed** (many places/time zones), tools like **Git/GitHub**, hence **documentation** **+ precise interfaces**. - **Interface** = *what* is accepted (inputs, format) and *what* is returned, **not how**. Example: inputs W, X, Y → output Z. **Code behind an interface can change** if the interface stays.

Testing. Check the software **behaves per requirements**; bugs still exist. Cost of bugs to the US economy: **~$59.5 billion (2002) → $1.1 trillion (2016)**.

	Who	Where	Goal
Alpha	**internal employees**	**lab / staging**	catch as many issues as possible **before public release**
Beta	**actual users**	**real-life setting**	real-world feedback

Maintenance. After release: **monitor** how users use it, **change code** for upgrades/fixes (missed conditions, browser-specific UI bugs), **add features**, plan the next release from feedback.

1.3 Development models



- **Waterfall:** phases strictly in sequence. Story: 2+3+4+3 = **12 months**, client sees it first at release: "I don't like it!". **Drawbacks:** (1) later change raises **cost and time**, (2) **clients don't know what they need**, (3) **designers don't know** which design is feasible/usable.

- **V-model:** waterfall **bent upward** after implementation. Left arm: concept → requirements/architecture → detailed design → **implementation** (bottom). Right arm: test/verify **individual modules** against design → verify/validate **whole system** against architecture → operation & maintenance.
- **Iterative:** many iterations: plan/build key features → deploy → **client feedback** → refine next iteration → repeat until complete.
- **Agile:** an iterative model; **small increments**, **prototypes** of key features released quickly. Frameworks: **Scrum, Kanban**. Practices: **sprints, test-driven development**. An **umbrella term / philosophy**, more than frameworks.
- Teams **choose and adapt** a model to their process and software type.

1.4 Software vs building (and car)

	Building	Software
Parts	**tangible**	**logical, abstract**
Change	hard	**adaptable**, anyone on the team can change it
Over time	**static**	**keeps changing** (Winamp → Saavn/Gaana)
Similar	blueprint first, requirement gathering, site visits / understanding end users, scale models	

MCQ rules, Week 1

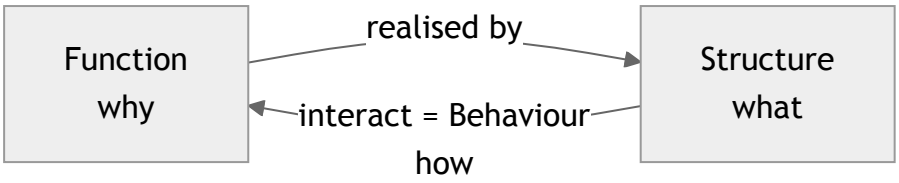
- "NOT a phase" → **UI design**. Stage before development → **Design**.
- Internal colleagues test → **alpha**; real users → **beta**.
- "Gather info/regulations" → requirements · "modules + interaction" → design · "feedback + next release" → maintenance · "release to select users, find errors" → testing.
- Lab builds a model for itself → **internal** client.
- Interface → what, not how; code **can** change. Waterfall's drawback → **changing requirements**.

Week 2: How to think about a design (FBS)

Phases and models don't say *how to think*. Experts use past systems as **reference points** and keep an **integrated view** asking **why, what, how**:

	Asks	Is	Mood-player example
Function (F)	**Why**	**Features**, deliberated from requirements	mood detection, playlist creation, user authentication
Structure (S)	**What**	**Physical + logical components** that realise functions	emotion camera, playlist algorithm, emotion tagging of songs
Behaviour (B)	**How**	**Interaction between structures** that achieves the function	camera recognises emotion → gives it to playlist generator → playlist by mood

Mood player: auto-detects mood, plays music by mood, secure authentication, remembers choices, recommends by mood + choice.



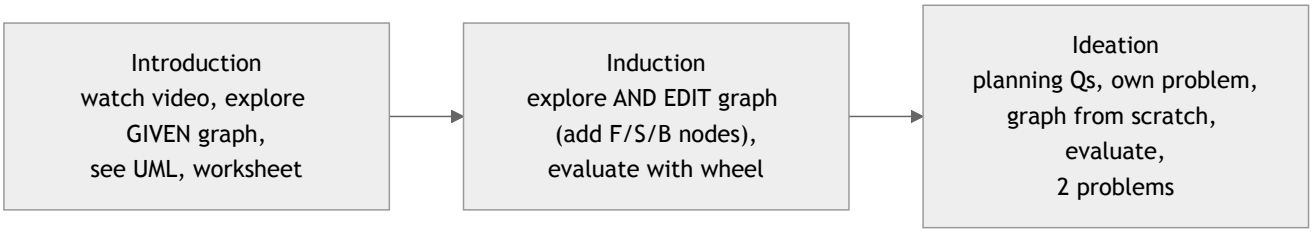
Why integrate F, B, S? - Not integrated → **incoherent/incomplete design, many solutions, undecidable** (mood detection alone: webcam vs voice vs history). - Integrated → (1) **unified view**, (2) **all requirements met**, (3) **team/stakeholder communication**, (4) **avoid getting stuck**.

How to integrate (5 steps): start from a **known solution** (e.g. existing mood cameras; check it fits) → think as **end user and system** to generate **behaviours** → **map** the structures working together in each behaviour → check how structures + behaviours **satisfy a requirement** → you now have F, B, S **identified and mapped**.

FBS graph: nodes = F/B/S (slides: F yellow, S blue, B red), links connect them, start from **any** node, **many valid graphs** (e.g. BSF, FSB), never one correct graph.

Think & Link (tool to build + evaluate FBS graphs)

Pedagogical agent (prompts), notes box, tabs unlock as tasks complete.



Introduction worksheet (6 tasks): list F/S/B nodes · true/false with reasons · describe F, S, B in own words · list **connector types** (F-S, S-B ...) · examples of **pairs and trios** · write abstract sentences like "Function is implemented by Structure used during Behaviour". **UML from FBS (component diagram):** one **component per structure node**; reflect **hierarchy** between structure nodes; check components that **implement no function**; every F-B-S link should have a matching interface.

Evaluation wheel (core: syntax, semantic, pragmatism). Each slice rated **Meets Expectation / Needs Improvement / Inadequate / Missing**, with a reason and planned changes. Can edit the graph and re-evaluate.

Criterion	Means
Complexity	**number of nodes** in each of F, B, S
Connectivity	F, S, B nodes **interconnected** (linking purpose, concept, component, mechanism)
Formal realisation	graph **converts to UML diagrams**
Level adjacency, Validity, Consistency, Levels	on the wheel; **not defined in lectures** (only names appear)

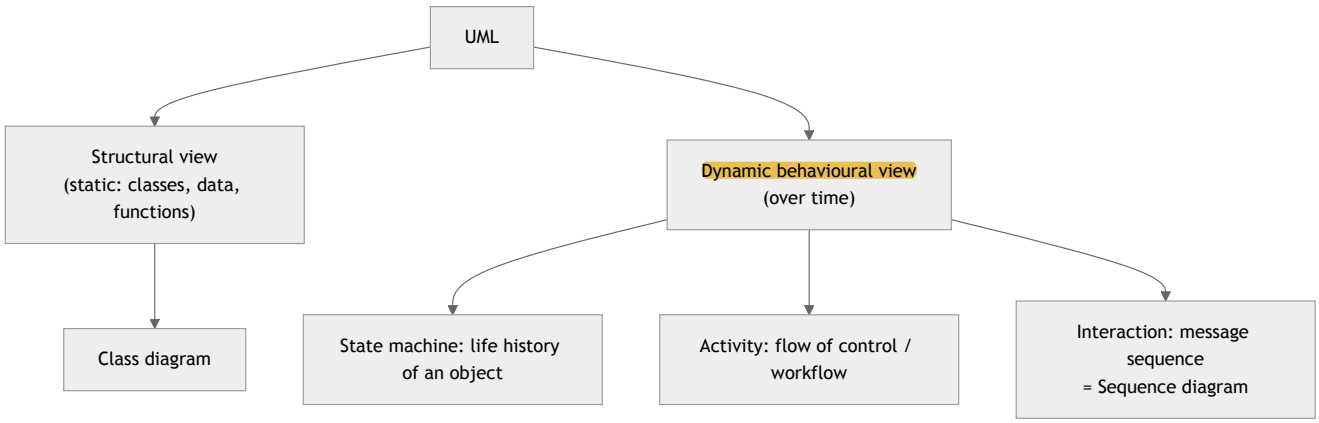
Summary (L13): conceptual design = a design description, **graphical, numerical or textual**; technique FBS, tool Think & Link.

MCQ rules, Week 2

- No integration → **incoherent/redundant, multiple solutions**. Integration advantages → **all four**.
- How many graphs → **multiple**. Reason to link → **integrated solution**.
- Explore **then edit** → **Induction**. Build from scratch → **Ideation**.
- "Camera → playlist generator → playlist" = **interaction/behaviour** (and implementation of the feature), **not a use case**.
- Camera, algorithm = **structure**; "mood detection" = **function**; "user places face near camera" / "system fetches songs" = **behaviour**.
- Complexity = **count**; connectivity = **links**; formal realisation = **UML**.

Week 3: Detail it in UML, then judge it

Modelling = creating an **external, explicit representation** of the system to be built. Benefits: plan what to code, **communicate to a larger team**, overview. FBS = **idea/concept**; UML = **more detail, multiple views**. (Example: one FBS function, mood detection, is expanded as a sequence diagram.)



Other UML diagrams exist (state chart, communication, use case, component, deployment...); a **subset** is used per design.

Class diagram (main building block of OO modelling): box with **3 compartments: name / attributes (data members) / operations (member functions)** + relationships. So it includes behaviour as operations. Mood player classes: User, User Voice Profile, Playlist, Song, User Mood (`moodName`, `detectMood()`). ATM: `balance` = data member, `setOption()` = member function.

Sequence diagram: objects from the class diagram on top, each with a **lifeline**; **horizontal arrows = messages in time order** (class-diagram functions such as `detectMood`, `generatePlaylist`). **One per requirement** (mood detection, login). Login has **alt/else**: valid → fetch profile/playlist → success; else → unsuccessful. For a check function returning false, read the **else message** ("Incorrect PIN").

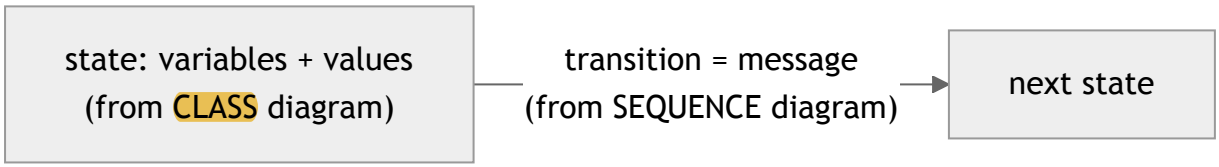
Understanding a design

Design comprehension = understanding a given design. Needed because developers work on **large existing systems** and add features (e.g. integrating the mood player into Amazon Music).

Student difficulties	Expert knowledge (4 kinds)
(1) **notations** (syntax)	**Problem domain** (how a music player works)
(2) mapping **requirements ↔ design**	**Design diagram** knowledge (syntax + semantics)
(3) **connecting diagrams** (sequence uses class-diagram objects)	**Main goals** (purpose of each diagram; made of **sub-goals**: input, success, failure)
	Dynamic behaviours (how class **data members change** as the sequence runs)

VeriSIM = Verifying designs by Simulating scenarios; gives an integrated understanding of **class + sequence** diagrams. Context: **automated door-locking system**: (1) if no passcode set, user **registers** one; (2) **lock** + correct passcode → locks, incorrect → **stays unlocked**; (3) **unlock** + correct → unlocks, incorrect → **stays locked**. Diagrams: class (Door, Controller, Passcode) + several sequence diagrams.

Design tracing = build a **state diagram for one scenario**:



e.g. start `lock=true`, E1 "set option unlock", E2 "enter correct passcode", E3 "door unlocks" → `lock=false`.

VeriSIM stages: Introduction (objectives) → **Problem understanding** (requirements + diagrams) → **Design tracing** (4 challenges, any order: **explore, correct, complete, construct the model**) → **Reflection**. Each activity has evaluation + reflection.

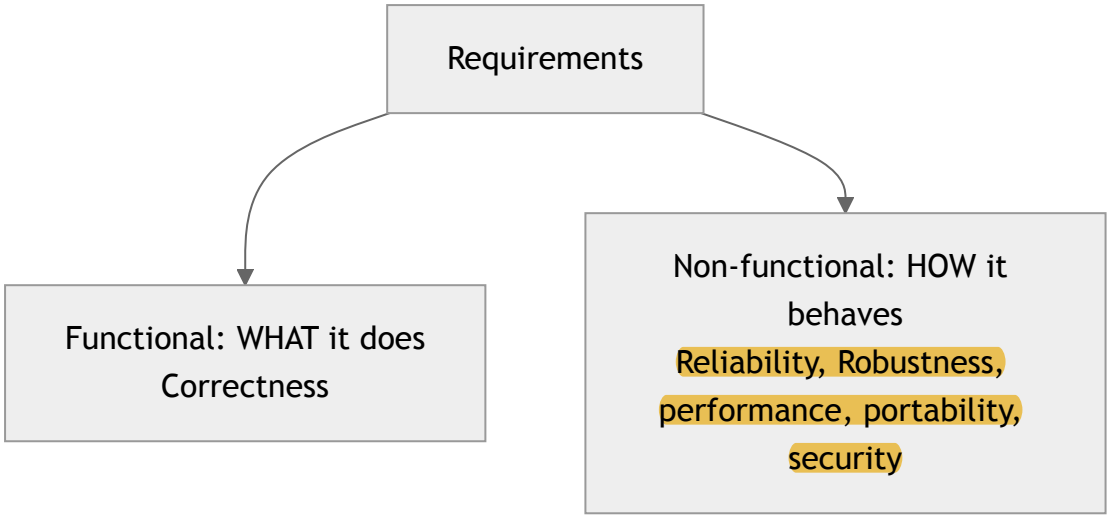
Domain Knowledge
Design Diagram Knowledge
Main Goals
Dynamic Behaviours

Verifying Designs by Simulating Scenarios

Automated door locking example

(design Tracing Strategy : Build state Diagram)

Judging a design: quality (tiffin-service analogy)



1 functional and 2 non-functional requirements!

Attribute	Definition	Tiffin	App
Correctness	behaves **according to specification** (a **functional** requirement)	on time, no mix-up (breakfast for lunch)	all requirements satisfied (as in VeriSIM tracing)
Reliability	behaves the **same way over time** in the same environment	misses >3 meals a week	does **not crash often**; withdraw/deposit work
Robustness	**recovers from errors / unexpected input**	chef sick, replacement	**payment failure**, negative/large numbers, high traffic

Checking reliability/robustness checks **how it behaves**, not the requirements. Quality attributes are **implicit, not in diagrams**, so **write them down** with the decisions based on them.

MCQ rules, Week 3

- Two views → **structural + dynamic behavioural**. Design tracing → **state diagram of a scenario**; state variables ← **class**, transitions ← **sequence**.
- Functional = **features**; non-functional = **behaviour**.
- Handles failures/unexpected input → **robustness**; crashes often → **reliability**.
- "Class diagram only has data members" → **False**. `deposit()` returns → **True/False**.

Week 4: Link it all (levels of detail)

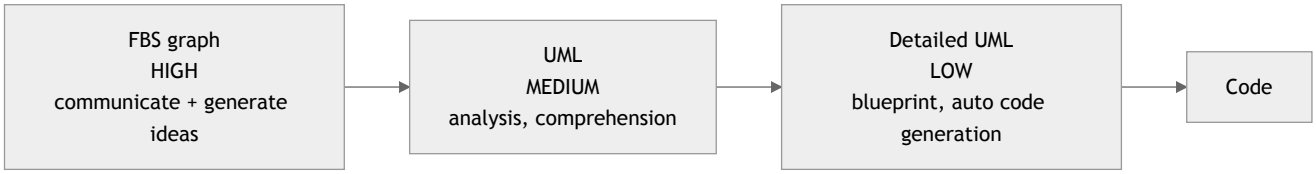
Recap: week 1 modules (**modularization**) and phases; weeks 2-3 models. Development is a **systematic problem-solving process**, phases are broad and done **iteratively in many ways**. Software design is harder than architecture because software is **abstract and intangible**, though design processes are similar across domains.

FBS definitions: F = functionalities/features, **S = physical + logical components**, B = how structures interact.

Mapping FBS to the process (table from the Rational Unified Process): linking two FBS elements triggers a process.

Link	Process initiated
Function ↔ Behaviour	**Requirement definition**
Structure ↔ Behaviour	**Design analysis**, testing, implementation, requirements change
Behaviour ↔ Behaviour, **Structure ↔ Structure**	**Evaluation, decomposition, refinement**
Function ↔ Structure	**Reusability and change in needs**

Abstraction = hiding details at each stage of design; **level of detail = level of abstraction**.



FBS → UML: **functions → use cases**, **structures → class diagrams**, **behaviours → sequence diagrams**. Detailed UML ↔ code: **each model class maps to one or more code classes**; tools **auto-generate code**.

Term	Meaning
Modularization	split into **independent co-working modules**, the system's **building blocks**
Abstraction	**hide details** at each design stage
Decomposition	**break down** implementation components

Experts spend a lot of time **early**: understanding the problem, building models, **checking models in different scenarios**, before implementing. That is **software conceptual design**.

MCQ rules, Week 4

- Structures in FBS → **physical and logical components**.
- Purposes of modelling (MSQ) → **communication/idea generation + analysis/comprehension + blueprint for implementation**; **not** "gather requirements from stakeholders".
- Communicate key ideas first → **FBS**; devs transform directly to code → **detailed UML**. FBS = **high-level overview**, UML = **more detailed**.
- Mapping questions: Function↔Behaviour = **requirements**; Structure↔Behaviour = **design analysis**; B↔B / S↔S = **evaluation/decomposition**; F↔S = **reusability**.
- Independent modules → modularization; hide details → abstraction; breaking components → decomposition.

Last 10 minutes

1. Requirements → Design → Development → Testing → Maintenance. Clients: external / internal / software.
2. Alpha = internal + lab; Beta = real users. Waterfall problem = change. Agile = Scrum, Kanban, sprints, TDD.

3. Interface = what not how; code can change.
4. F = why/features, S = what/components, B = how/interactions. No integration → incoherent, many solutions.
5. Many FBS graphs. Induction edits, Ideation creates. Complexity = nodes, Connectivity = links, Formal realisation = UML.
6. UML: structural (class: name/attributes/operations) + dynamic (sequence: lifelines, messages). VeriSIM tracing = state diagram of a scenario.
7. Correctness (functional) · Reliability (crashes) · Robustness (failures, odd input).
8. FBS (high) → UML (medium) → detailed UML (low) → code. Modularization / abstraction / decomposition.